

YASH KUMAR

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology (NIT), Jalandhar, India **2024 – 2026**

M.Tech, Data Science & Engineering | Advisors: Dr. Amritpal Singh, Dr. Himanshu Buckchash

Thesis: "Application of Instance vs. Amodal Deep Models for Mitochondria Segmentation Using Simulated and Real Confocal Microscopy Data."

Punjabi University, Patiala, India

2019 – 2023

B.Tech, Computer Science & Engineering

HONORS & QUALIFICATIONS

- **GATE 2026** — Qualified, Computer Science & Information Technology (CS/IT). *GATE (Graduate Aptitude Test in Engineering) is a national, government-administered exam used across India for admission to graduate (Master's/Ph.D.) engineering programs and for public-sector technical recruitment; it is highly competitive, with a single-digit qualifying percentage nationally.*
- **GATE 2024** — Qualified, Data Science & Artificial Intelligence (DA).
- **UGC NET** — Qualified for Assistant Professor & Ph.D. Admission, June 2026 (89.53 percentile). *UGC NET (National Eligibility Test) is India's national qualifying examination for eligibility to serve as a university Assistant Professor and/or to receive government doctoral fellowships (JRF).*

PUBLICATIONS

- [1] Y. Kumar, A. Singh, H. Buckchash, "From Visual Plausibility to Physical Grounding: A Review of Modern Amodal Perception." Accepted — International Conference on Computing, Communication and Artificial Intelligence (IC3AI).
- [2] Y. Kumar, H. Buckchash, A. Singh, "Application of Instance vs. Amodal Deep Models for Mitochondria Segmentation Using Simulated and Real Confocal Microscopy Data." In preparation — Biomedical Signal Processing and Control (Elsevier).
- [3] Y. Kumar, H. Buckchash, A. Singh, "Amodal Computer Vision: A Systematic Review of the Techniques and Applications Over a Decade." Submitted — Computer Science Review (Elsevier).

RESEARCH EXPERIENCE

Graduate Researcher — NIT Jalandhar

2024 – 2026

- Designed and executed a systematic benchmark comparing four architectures (Mask2Former, AISFormer, DETR, classical baseline Nellie) for mitochondria segmentation on physics-simulated confocal microscopy data with paired visible/amodal ground truth; best model (Mask2Former) reached test mIoU = 0.786, while the amodal model AISFormer led on boundary accuracy (HD95 = 5.23 px) at the cost of detection AP (0.265 vs. 0.518).
- Built and debugged full training/evaluation pipelines for AISFormer (Detectron2) and Mask2Former (HuggingFace Transformers) from scratch, and engineered a PBS-managed HPC training infrastructure on H100 MIG GPUs to support reliable long-running (70,000-iteration) jobs.
- Evaluated zero-shot generalization of all four benchmarked models to real confocal microscopy acquisitions, characterizing distinct sim-to-real failure modes per architecture.
- Conducted a systematic literature review of 100+ amodal computer vision papers (mini-PRISMA methodology, 368 records screened to 100) and produced a hierarchical taxonomy of the field.

TEACHING EXPERIENCE

REVA University, Bengaluru, India

- Subjects: Principles of AI; Data Visualization Techniques

INTERNSHIP EXPERIENCE

Data Science Engineer Intern — HopingMinds, Mohali, India

2023

- Worked on applied data science and machine learning tasks as part of the engineering team, contributing to model development and data pipeline workflows during the internship.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL | **ML/DL:** PyTorch, HuggingFace Transformers, Detectron2, TensorFlow

Infrastructure: PBS/HPC scheduling, CUDA/MIG GPU management, conda, Git | **Other:** LaTeX, Jupyter, AI-assisted workflows (Claude, Gemini)

Competitive Programming: Regular problem solver on LeetCode, CodeChef, GeeksforGeeks — consolidated profile on Codolio (codolio.com/profile/Sd0CVmbL)

REFERENCES

Dr. Amritpal Singh — Assistant Professor, Dept. of CSE, NIT Jalandhar — apsingh@nitj.ac.in

Dr. Himanshu Buckchash — Assoc. Prof. (FH), IMC Krems University of Applied Sciences — hbuckchash@cs.iitr.ac.in